

Lars W. Osterberg

Department of Mechanical Engineering · Stanford University

[Email](#) · [Personal Website](#)

Education

Stanford University

Stanford, CA

Master of Science, Mechanical Engineering

Sept 2025 – Present

- **GPA:** 4.03 / 4.00
- **Relevant Coursework:** Robot Perception, Principles of Robot Autonomy, Linear Algebra (SVD), State Estimation, Convex Optimization, Optimal & Learning-Based Control.

University of California, San Diego

La Jolla, CA

Bachelor of Science, Mechanical Engineering (Cum Laude)

Sept 2021 – June 2025

- **GPA:** 3.90 / 4.00 (**Major GPA:** 3.96 / 4.00)
- **Research Groups:** Poulikakos Lab, Coordinated Robotics Laboratory

Publications & Presentations

Preprints & Manuscripts Under Review

1. **Osterberg, L.**, Wang, M., & Schwager, M. (2026). UniMem: Unifying Multimodal Memory and Control for Vision-Language-Action Models. *arXiv preprint arXiv:2608.22869*. [arXiv:2608.22869](#). (Under review at ICRA 2027) [[Website](#)] [[Code](#)]
2. Wang, M., **Osterberg, L.**, Tian, S., Shorinwa, O., Wu, J., & Schwager, M. (2026). InSight: Self-Guided Skill Acquisition via Steerable VLAs. *arXiv preprint arXiv:2606.24884*. [arXiv:2606.24884](#). (Under review at CoRL 2026) [[Website](#)] [[Code](#)]
3. Kim, H., Chen, T., Sun, J., **Osterberg, L. W.**, Chen, Q., Wang, K., & Schwager, M. (2026). LEGS: Fine-Tuning Teleop-Free VLAs for Humanoid Loco-manipulation in an Embodied Gaussian Splatting World. *arXiv preprint arXiv:2606.01458*. [arXiv:2606.01458](#). (Under review at CoRL 2026) [[Website](#)]

Peer-Reviewed Journal and Conference Articles

1. Miranda, M. H. I., **Osterberg, L. W.**, Shah, D. H., Regulagadda, K., & Poulikakos, L. V. (2025). 3D-architected gratings for polarization-sensitive, nature-inspired structural color. *Nanophotonics*, 14(5), 547–558. [doi:10.1515/nanoph-2024-0657](#)

Conference Proceedings & Oral Presentations

1. Poulikakos, L. V., Holland, J., Miranda, M. I., **Osterberg, L.**, Kim, C., Chattopadhyay, S., & Boechler, N. (2025). Tailoring wave-matter interactions with Mie-resonant and acoustoplasmonic metasurfaces. In *Plasmonics: Design, Materials, Fabrication, Characterization, and Applications XXIII* (Vol. PC13579, p. PC1357906). SPIE. [doi:10.1117/12.3065158](#)
2. Poulikakos, L. V., Kirya, P., Haddadin, Z., Chattopadhyay, S., Maddirala, G., Miranda, M. I., **Osterberg, L.**, Mestre-Farrera, A., & Yang, J. (2025). Nature-inspired photonic surfaces for next-generation imaging and diagnostics. In *Metamaterials, Metadevices, and Metasystems 2025* (Vol. PC13577, p. PC135770A). SPIE. [doi:10.1117/12.3065137](#)
3. Miranda, M. H. I., Phillips, I. R., **Osterberg, L. W.**, Garay, J. E., & Poulikakos, L. V. (2025). Laser processing approaches for functional nano- and microstructures with photonic applications. In *Plasmonics: Design, Materials, Fabrication, Characterization, and Applications XXIII* (Vol. PC13579, p. PC1357912). SPIE. [doi:10.1117/12.3064729](#)
4. Poulikakos, L. V., Miranda, M. I., **Osterberg, L.**, Kirya, P., & Phillips, L. (2024). Rationally designed colorimetric metasurfaces for healthcare applications. In *Active Photonic Platforms (APP) 2024* (Vol. PC13110, p. PC1311012). SPIE. [doi:10.1117/12.3027892](#)

5. **Osterberg, L. W.** (2024, August). *3D-Architected Gratings for Structural Color Applications*. Oral Presentation at the UC San Diego Undergraduate Research Conference, La Jolla, CA. [[Program Abstract](#), p. 229]

Research Experience

Graduate Research Assistant

December 2025 – Present

Multi-Robot Systems Lab (MSL), Stanford University

Stanford, CA

Advisor: Dr. Mac Schwager

- Designed and implemented **UniMem**, a VLA architecture unifying multimodal spatial-semantic memory with real-time control, improving long-horizon, non-Markovian task success rates by 36.5% over hierarchical-memory VLA baselines.
- Formulated a memory caching mechanism for efficient, lightweight access to historical information, enabling a 6× latency reduction over hierarchical baselines.
- Deployed and benchmarked VLA policies on xArm physical hardware to support several lab projects.

Undergraduate Research Assistant

Oct 2023 – May 2025

Poulikakos Lab, UC San Diego

La Jolla, CA

Advisor: Dr. Lisa V. Poulikakos

- Designed colorimetric metasurfaces and 3D-architected gratings in CAD, fabricated via two-photon lithography for diagnostic and photonic applications.
- Formulated an analytical diffraction model linking experimental AFM and spectroscopy data with optical theory, contributing to two peer-reviewed publications.

Selected Projects

Acoustic Localization Robot

March 2024 – Feb 2025

UCSD Coordinated Robotics Laboratory

La Jolla, CA

- Engineered hardware and software systems for a self-localizing robotic module using a Single Board Computer (SBC).
- Implemented a real-time C/C++ cross-correlation algorithm using open-source software (FFTW) for microsecond-precision acoustic signal detection.
- Programmed the Programmable Real-Time Unit (PRU) with open-source I/O driver libraries to achieve a 200 kHz ADC sampling rate.

Automated Graphene Transfer System

Nov 2024 – March 2025

UCSD Department of Mechanical and Aerospace Engineering

La Jolla, CA

- Designed and built a microcontroller-driven automation system to eliminate human error in transferring nm-thin graphene onto silicon substrates.
- Integrated real-time conductivity probing, liquid-level sensing, solenoid drainage, and pump actuation to automate multi-bath chemical cleaning and drying cycles. [[Video](#)] [[Website](#)]

Honors, Service & Leadership

Stanford University Environmental Health & Safety Committee Member

Sept 2025 –

Present

- Representing the graduate student body on campus-wide safety initiatives.

Woodruff School Chair's Interdisciplinary Graduate Fellowship

2025

Georgia Institute of Technology

Presidential Fellowship Award

2025

Georgia Institute of Technology